

## Statistics for Data Science - 2

### Week 7 Graded assignment

1. Let  $X_1, X_2, X_3$  are three independent and identically distributed random variables with mean  $\mu$  and variance  $\sigma^2$ . Given below are 3 different formulations of sample mean. (Observe that  $E[A] = E[B] = E[C]$ ).

$$A = \frac{X_1 + X_2 + X_3}{3}$$

$$B = 0.1X_1 + 0.3X_2 + 0.6X_3$$

$$C = 0.2X_1 + 0.3X_2 + 0.5X_3$$

Choose the correct option from the following:

- (a)  $\text{Var}(A) = \text{Var}(B) = \text{Var}(C)$
- (b)  $\text{Var}(A) \geq \text{Var}(B) \geq \text{Var}(C)$
- (c)  $\text{Var}(A) \leq \text{Var}(B) \leq \text{Var}(C)$
- (d)  $\text{Var}(A) \leq \text{Var}(C) \leq \text{Var}(B)$

**Solution:**

Let  $X_1, X_2, X_3 \sim \text{i.i.d.} X$ , where  $E[X] = \mu$ ,  $\text{Var}(X) = \sigma^2$

$$\begin{aligned}\text{Var}(A) &= \text{Var}\left(\frac{X_1 + X_2 + X_3}{3}\right) \\ &= \frac{1}{9} (\text{Var}[X_1] + \text{Var}[X_2] + \text{Var}[X_3]) \\ &= \frac{1}{9} (3\sigma^2) = \frac{\sigma^2}{3}\end{aligned}$$

$$\begin{aligned}\text{Var}(B) &= \text{Var}(0.1X_1 + 0.3X_2 + 0.6X_3) \\ &= 0.01\text{Var}[X_1] + 0.09\text{Var}[X_2] + 0.36\text{Var}[X_3] \\ &= 0.46(3\sigma^2) \\ &= 1.38\sigma^2\end{aligned}$$

$$\begin{aligned}\text{Var}(C) &= \text{Var}(0.2X_1 + 0.3X_2 + 0.5X_3) \\ &= 0.04\text{Var}[X_1] + 0.09\text{Var}[X_2] + 0.25\text{Var}[X_3] \\ &= 0.38(3\sigma^2) \\ &= 1.14\sigma^2\end{aligned}$$

Therefore,  $\text{Var}(B) \geq \text{Var}(C) \geq \text{Var}(A)$ .

2. A random sample of size 25 is collected from a normal population with mean of 50 and standard deviation of 5. Find the variance of the sample mean.

**Solution:**

We know that variance of the sample mean  $\bar{X}$  is given by

$$\begin{aligned}\text{Var}[\bar{X}] &= \frac{\sigma^2}{n} \\ &= \frac{5^2}{25} = 1\end{aligned}$$

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3. A fair die is rolled 100 times. Let  $X$  denote the number of times six is obtained. Find a bound for the probability that  $\frac{X}{100}$  differs from  $\frac{1}{6}$  by less than 0.1 using weak law of large numbers.

- (a) at least  $\frac{5}{36}$
- (b) at least  $\frac{31}{36}$
- (c) at most  $\frac{5}{36}$
- (d) at most  $\frac{31}{36}$

**Solution:**

$X$  denotes the number of times six is obtained on rolling a fair die 100 times. Let  $X_1, X_2, \dots, X_{100}$  be 100 i.i.d. samples such that

$$X_i = \begin{cases} 1 & \text{if six appears on rolling a fair die} \\ 0 & \text{otherwise} \end{cases}$$

$$E[X_i] = \mu = \frac{1}{6} \text{ and}$$

$$\text{Var}(X_i) = \sigma^2 = \frac{5}{36}$$

Notice that  $X = X_1 + X_2 + X_3 + \dots + X_{100}$

To find: Bound on  $P\left(\left|\frac{X}{100} - \frac{1}{6}\right| < 0.1\right)$ .

By weak law of large numbers, we have

$$\begin{aligned} P(|\bar{X} - \mu| < \delta) &\geq 1 - \frac{\sigma^2}{n\delta^2} \\ \Rightarrow P\left(\left|\frac{X}{100} - \frac{1}{6}\right| < 0.1\right) &\geq 1 - \frac{5}{36 \times 100 \times 0.01} \\ \Rightarrow P\left(\left|\frac{X}{100} - \frac{1}{6}\right| < 0.1\right) &\geq 1 - \frac{5}{36} = \frac{31}{36} \end{aligned}$$

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4. Let  $X_1, X_2, \dots, X_5$  be i.i.d. samples whose distribution has mean 20 and variance 4. Suppose the sample variance is defined as

$$S^2 = \frac{(X_1 - \bar{X})^2 + \dots + (X_5 - \bar{X})^2}{5}$$

, where  $\bar{X} = \frac{X_1 + X_2 + \dots + X_5}{5}$ . Find the expected value of  $S^2$ .

**Solution:**

$$E[\bar{X}] = \mu = 20 \text{ and } \text{Var}[\bar{X}] = \frac{\sigma^2}{n} = \frac{4}{5} = 0.8.$$

$$\begin{aligned}
E[S^2] &= E \left[ \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n} \right] \\
&= \frac{1}{n} E \left[ \sum_{i=1}^n (X_i - \bar{X})^2 \right] \\
&= \frac{1}{n} E \left[ \sum_{i=1}^n (X_i^2 + \bar{X}^2 - 2X_i\bar{X}) \right] \\
&= \frac{1}{n} E \left[ \sum_{i=1}^n X_i^2 + n\bar{X}^2 - 2n\bar{X} \sum_{i=1}^n X_i \right] \\
&= \frac{1}{n} E \left[ \sum_{i=1}^n X_i^2 + n\bar{X}^2 - 2n\bar{X}^2 \right] \\
&= \frac{1}{n} E \left[ \sum_{i=1}^n X_i^2 - n\bar{X}^2 \right] \\
&= \frac{1}{n} \left[ \sum_{i=1}^n E[X_i^2] - nE[\bar{X}^2] \right] \\
&= \frac{1}{n} \left[ \sum_{i=1}^n (\sigma^2 + \mu^2) - n \left( \frac{\sigma^2}{n} + \mu^2 \right) \right] \\
&= \frac{1}{n} [(n\sigma^2 + n\mu^2) - (\sigma^2 + n\mu^2)] \\
&= \frac{(n-1)\sigma^2}{n}
\end{aligned}$$

Here,  $n = 5$ , therefore,  $E[S^2] = \frac{4}{5} \times 4 = 3.2$

5. Suppose  $X_i \sim \text{Normal} \left( 0, \frac{1}{i^2} \right)$ , where  $i = 1, 2, \dots, 9$  and  $X_1, X_2, \dots, X_9$  are independent to each other. Let  $Y$  be a random variable defined as  $Y = \sum_{i=1}^9 iX_i$ . Find the variance of  $Y$ .

Answer: 9

**Solution:**

$$\begin{aligned}\text{Var}(Y) &= \text{Var}\left(\sum_{i=1}^9 iX_i\right) \\ &= \text{Var}(X_1 + 2X_2 + 3X_3 + \dots + 9X_9) \\ &= \text{Var}(X_1) + \text{Var}(2X_2) + \dots + \text{Var}(9X_9) \\ &= \text{Var}(X_1) + 4\text{Var}(X_2) + \dots + 81\text{Var}(X_9) \\ &= \frac{1}{1^2} + 4\left(\frac{1}{2^2}\right) + \dots + 81\left(\frac{1}{9^2}\right) \\ &= 9\end{aligned}$$

6. A random sample of size 50 is collected from a population  $P$ , where  $P \sim \text{Uniform}[0,12]$ . Find a lower bound on the probability that the sample mean will be at most 3 units away from the actual mean using the weak law of large numbers.

**Answer:** 0.733

**Solution:**

$$P \sim \text{Uniform}[0, 12]$$

$$E[P] = \mu = \frac{0 + 12}{2} = 6, \text{Var}(P) = \sigma^2 = \frac{(12 - 0)^2}{12} = \frac{144}{12} = 12$$

By weak law of large numbers, we have

$$\begin{aligned}P(|\bar{X} - \mu| < \delta) &\geq 1 - \frac{\sigma^2}{n\delta^2} \\ \Rightarrow P(|\bar{X} - \mu| < 3) &\geq 1 - \frac{12}{50 \times 9} = \frac{73}{75} = 0.9733\end{aligned}$$

7. Suppose a random sample is used to estimate the proportion of voters in a city. If the sample proportion is roughly 0.45, what sample size is necessary so that the standard deviation of the sample proportion is 0.02?

**Answer:** 619

**Solution:**

Let the random variable  $X$  represents that the selected candidate is a voter.

Let  $X_i$  be defined as

$$X_i = \begin{cases} 1, & \text{if the selected candidate is a voter} \\ 0, & \text{otherwise} \end{cases}$$

Define an event  $A$  as  $A : X = 1$ .

It is given that  $P(A) = 0.45$ .

We know that  $\text{Var}(S(A)) = \frac{P(A)(1 - P(A))}{n}$

$$\sqrt{\frac{p(1-p)}{n}} = 0.02 \implies \sqrt{\frac{(0.45)(0.55)}{n}} = 0.02 \implies n = 618.75 \approx 619$$

8. The average life (in years) of an electronic watch follows an exponential distribution with parameter  $\frac{1}{2}$ . Find the lower bound on the probability that the mean life of a random sample of 50 such watches falls between 1 and 3 years. Enter your answer correct to two decimals.

Hint: Use weak law of large numbers.

Answer: 0.92

**Solution:**

Let the random variable  $X$  represents the life of an electronic watch.

It is given that  $X \sim \text{Exp}(1/2)$  and 50 such samples are taken.

$$E[X] = \mu = 2, \text{Var}(X) = \sigma^2 = 4$$

To find: a lower bound on  $P(1 < \bar{X} < 3)$ .

By weak law of large numbers, we have

$$\begin{aligned} P(|\bar{X} - \mu| < \delta) &\geq 1 - \frac{\sigma^2}{n\delta^2} \\ \Rightarrow P(|\bar{X} - 2| < 1) &\geq 1 - \frac{4}{50 \times 1} = \frac{23}{25} = 0.92 \end{aligned}$$